

Mathematical Methods I

Curves in $\mathbb{E}^3$

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1. Curves in Euclidean space

A curve describes the motion of a point in three-dimensional Euclidean space. The parameter t can be interpreted as time, and $\alpha(t)$ gives the position of the point at that instant.

Definition 1.1 (Parametrized curve)

A **curve** in $\mathbb{E}^3$ is a differentiable function

$$\alpha : I \longrightarrow \mathbb{E}^3, \quad t \longmapsto \alpha(t),$$

where $I \subseteq \mathbb{R}$ is an open interval. Its **coordinate functions** are the differentiable real-valued functions $\alpha_1, \alpha_2, \alpha_3$ such that

$$\alpha(t) = (\alpha_1(t), \alpha_2(t), \alpha_3(t)).$$

Example 1.2 (The line)

Given a point p and a constant vector $q \neq 0$, the curve

$$\alpha(t) = p + tq, \quad t \in \mathbb{R},$$

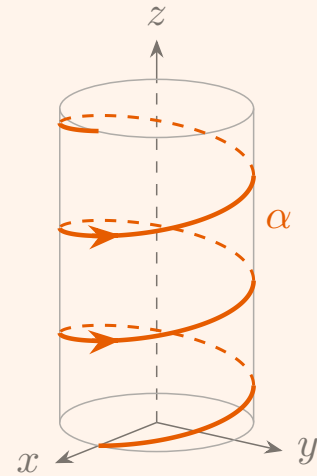
parametrizes the line passing through $p = \alpha(0)$ with direction q . If $q = 0$, the same expression defines a constant curve.

Example 1.3 (The circular helix)

For $a > 0$ and $b \neq 0$,

$$\alpha(t) = (a \cos t, a \sin t, bt), \quad t \in \mathbb{R},$$

revolves around the z -axis on the cylinder $x^2 + y^2 = a^2$. Its projection onto the xy -plane is a circle of radius a . Its height changes at the constant rate b : it rises if $b > 0$ and descends if $b < 0$.



Circular helix ($b > 0$).

2. The velocity vector

The derivative of a curve measures how its position changes with the parameter. It is computed by differentiating each coordinate function and is interpreted as a vector based at the corresponding point of the curve.

Definition 2.1 (Velocity of a curve)

For a differentiable curve $\alpha : I \rightarrow \mathbb{E}^3$, its **velocity vector** at time t is

$$\alpha'(t) = \left(\frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t) \right)_{\alpha(t)}. \quad (1)$$

The subscript $\alpha(t)$ indicates the point at which the vector is based.

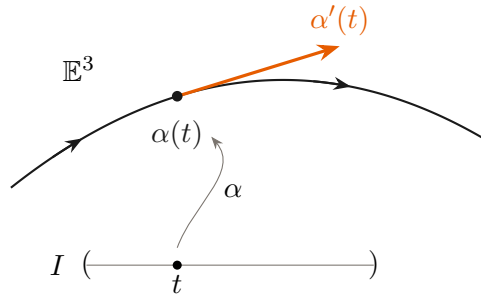


Figure 1: The parameter $t \in I$ determines the point $\alpha(t)$. The vector $\alpha'(t)$ is based at that point and is tangent to the curve whenever it is nonzero.

Tangent direction and speed

When $\alpha'(t) \neq 0$, its direction determines the tangent line, and its orientation corresponds to the motion as t increases. Its norm is the **speed**:

$$\|\alpha'(t)\| = \sqrt{(\alpha'_1(t))^2 + (\alpha'_2(t))^2 + (\alpha'_3(t))^2}.$$

Example 2.2 (Velocity in the preceding examples)

For the line $\alpha(t) = p + tq$, we have $\alpha'(t) = q_{\alpha(t)}$. For the helix,

$$\alpha'(t) = (-a \sin t, a \cos t, b)_{\alpha(t)}, \quad \|\alpha'(t)\| = \sqrt{a^2 + b^2}.$$

Both curves have constant speed.

3. From the secant to instantaneous velocity

For $t, t + \Delta t \in I$, the displacement

$$\alpha(t + \Delta t) - \alpha(t)$$

runs from $\alpha(t)$ to $\alpha(t + \Delta t)$. Dividing by $\Delta t \neq 0$ gives the **average velocity**:

$$\frac{\alpha(t + \Delta t) - \alpha(t)}{\Delta t}. \quad (2)$$

This vector is parallel to the secant line. For $\Delta t > 0$, it has the same orientation as the displacement; for $\Delta t < 0$, the orientation is reversed.

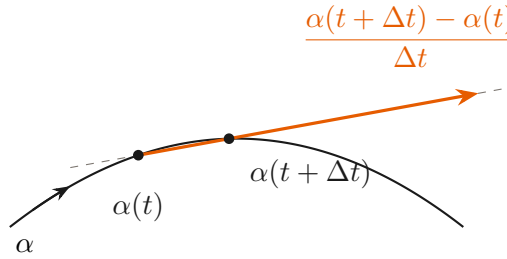


Figure 2: Difference quotient based at $\alpha(t)$ for a positive increment. As $\Delta t \rightarrow 0$, it converges to the velocity vector. The orange arrow follows the secant direction; its length is drawn to scale.

Proposition 3.1 (Definition by a limit)

Identifying displacements with vectors in $\mathbb{R}^3$,

$$\alpha'(t) = \left(\lim_{\Delta t \rightarrow 0} \frac{\alpha(t + \Delta t) - \alpha(t)}{\Delta t} \right)_{\alpha(t)}. \quad (3)$$

The limit is computed componentwise. Indeed, for $i = 1, 2, 3$,

$$\lim_{\Delta t \rightarrow 0} \frac{\alpha_i(t + \Delta t) - \alpha_i(t)}{\Delta t} = \alpha'_i(t).$$

Therefore,

$$\alpha'(t) = (\alpha'_1(t), \alpha'_2(t), \alpha'_3(t))_{\alpha(t)}.$$

If $\alpha'(t) \neq 0$, the secant directions approach the tangent direction. The tangent line at that instant is given by

$$\ell(u) = \alpha(t) + u(\alpha'_1(t), \alpha'_2(t), \alpha'_3(t)), \quad u \in \mathbb{R}.$$

If the velocity vector is zero, this formula does not determine a tangent direction.

4. Expression in the natural frame

In Cartesian coordinates, the vector fields U_1, U_2, U_3 assign the vectors of the standard basis to each point:

$$U_1(p) = (1, 0, 0)_p, \quad U_2(p) = (0, 1, 0)_p, \quad U_3(p) = (0, 0, 1)_p.$$

Every tangent vector can be written as $(v_1, v_2, v_3)_p = \sum_{i=1}^3 v_i U_i(p)$.

Proposition 4.1 (Components of velocity)

Applying this identity at $p = \alpha(t)$ gives

$$\alpha'(t) = \sum_{i=1}^3 \frac{d\alpha_i}{dt}(t) U_i(\alpha(t)). \quad (4)$$

That is,

$$\alpha'(t) = \alpha'_1(t)U_1(\alpha(t)) + \alpha'_2(t)U_2(\alpha(t)) + \alpha'_3(t)U_3(\alpha(t)).$$

All vectors on the right-hand side are based at the same point.

5. Action of velocity on a function

Let $f : \Omega \rightarrow \mathbb{R}$ be differentiable, where $\Omega \subseteq \mathbb{E}^3$ is open and $\alpha(I) \subseteq \Omega$. The value $f(\alpha(t))$ describes how f changes along the curve.

Theorem 5.1 (Differentiation along a curve)

The velocity vector acts on f as a directional derivative:

$$\alpha'(t)[f] = \frac{d}{dt}(f(\alpha(t))). \quad (5)$$

By the chain rule,

$$\alpha'(t)[f] = \sum_{i=1}^3 \frac{d\alpha_i}{dt}(t) \frac{\partial f}{\partial x_i}(\alpha(t)) \quad (6)$$

$$= \nabla f(\alpha(t)) \cdot \alpha'(t). \quad (7)$$

The dot product uses the Cartesian components of the velocity vector.

This expression measures the instantaneous rate of change of f with respect to the parameter t . The velocity vector is not normalized: its magnitude also affects the rate of change.

6. Reparametrization of curves

A change of parameter makes it possible to describe the points of a curve using a new variable.

Definition 6.1 (Change of parameter)

Let $\alpha : I \rightarrow \mathbb{E}^3$ and $h : J \rightarrow I$ be differentiable. The composite curve

$$\beta = \alpha \circ h : J \rightarrow \mathbb{E}^3, \quad \beta(s) = \alpha(h(s)),$$

is the reparametrization of α by h , in the broad sense used here. The new parameter s corresponds to the value $t = h(s)$ of the original parameter.

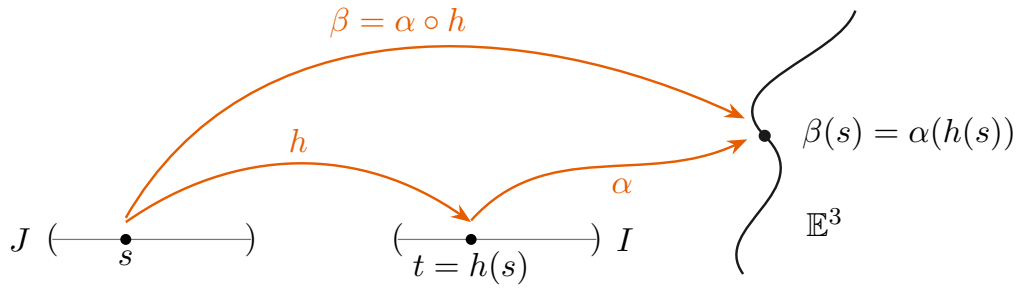


Figure 3: The map h sends $s \in J$ to $t = h(s) \in I$. The curves β and α reach the same point: $\beta(s) = \alpha(h(s))$.

The image satisfies $\beta(J) = \alpha(h(J))$. If $h(J) = I$, both curves trace the same entire path. For a regular change of parameter, one usually requires h to be bijective and $h'(s) \neq 0$.

Theorem 6.2 (Velocity of the reparametrized curve)

$$\boxed{\beta'(s) = h'(s) \alpha'(h(s))}. \quad (8)$$

Both sides are vectors based at $\beta(s) = \alpha(h(s))$.

Proof

We write $\beta(s) = (\alpha_1(h(s)), \alpha_2(h(s)), \alpha_3(h(s)))$. The chain rule for each component gives

$$\frac{d}{ds} \alpha_i(h(s)) = \alpha'_i(h(s)) h'(s), \quad i = 1, 2, 3.$$

Therefore,

$$\begin{aligned} \beta'(s) &= (h'(s) \alpha'_1(h(s)), h'(s) \alpha'_2(h(s)), h'(s) \alpha'_3(h(s)))_{\beta(s)} \\ &= h'(s) (\alpha'_1(h(s)), \alpha'_2(h(s)), \alpha'_3(h(s)))_{\alpha(h(s))} \\ &= h'(s) \alpha'(h(s)). \end{aligned}$$

Thus, $\|\beta'(s)\| = |h'(s)| \|\alpha'(h(s))\|$. The sign of h' preserves or reverses the orientation of the traversal when it is positive or negative, respectively.

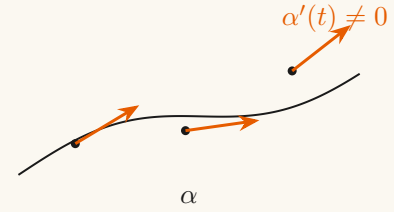
7. Regularity, periodicity, and path

Definition 7.1 (Regular curve)

A curve $\alpha : I \rightarrow \mathbb{E}^3$ is **regular** if its velocity vector never vanishes:

$$\alpha'(t) \neq 0 \quad \text{for every } t \in I.$$

At every instant, it has a well-defined tangent direction.



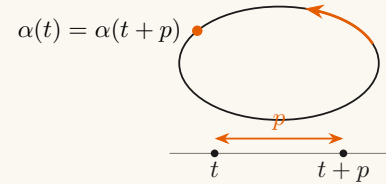
A nonzero tangent vector at every point.

Definition 7.2 (Periodic curve)

A curve $\alpha : \mathbb{R} \rightarrow \mathbb{E}^3$ is **periodic** if there is a number $p > 0$ such that

$$\alpha(t + p) = \alpha(t) \quad \text{for every } t \in \mathbb{R}.$$

The number p is a period: the parametrization repeats its positions after that interval.



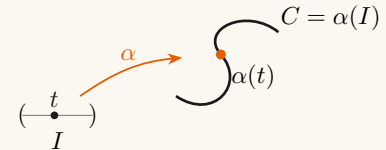
After one period, the same point is reached.

Definition 7.3 (Path or trace)

The **path** of α is the set of points it traces:

$$C = \alpha(I) = \{\alpha(t) : t \in I\} \subseteq \mathbb{E}^3.$$

The curve is a parametrized function; its path is a subset of space.



The map α traces the point set C .

Closed and open paths

To discuss endpoints, consider a continuous parametrization on a closed interval $[a, b]$ that is differentiable in its interior. It is **closed** if $\alpha(a) = \alpha(b)$. A periodic curve determines a closed traversal when restricted to any interval whose length equals one period.

If $\alpha(a) \neq \alpha(b)$, the traversal is not closed. One also considers open traversals that extend indefinitely. This terminology does not mean that the image is an open set in the topological sense.

Clarification: an open traversal may intersect itself

The fact that a curve is not closed does not imply that it is injective. The condition

$$\alpha(t_1) = \alpha(t_2) \implies t_1 = t_2$$

is an additional property that prevents points from being repeated. Likewise, having matching endpoints is not enough to guarantee a differentiable periodic extension: the compatibility conditions at the join must also be satisfied.